

DANG VAN TRONG

Email: van_trong.dang.ve6@naist.ac.jp
trong.dv211234M@sis.hust.edu.vn

Birth: Aug. 17, 1998

Address: Nara Institute of
Science and Technology, Japan

ResearchGate: researchgate.net/profile/Van-Trong-Dang

Google Scholar: scholar.google.com/citations?user=u_Eb52QAAAAJ

ORCID: orcid.org/0000-0002-9099-9334

EDUCATION

Nara Institute of Science and Technology, Japan - Ph.D. in Information Science (Expected Graduation: Sep 2026) (Oct 2023 – now)

Hanoi University of Science and Technology, Vietnam - M.Sc. in Control Engineering and Automation (GPA: 3.89/4.0 - Top 1% of University) (Apr 2021 – Oct 2022)

Hanoi University of Science and Technology, Vietnam - B.E. in Control Engineering and Automation (GPA: 3.60/4.0 - Top 1% of University) (Oct 2016 – Jan 2021)

RESEARCH EXPERIENCE

Cooperative Frameworks Using Conflict-aware Reference Generator and Semi-passive Control for Safe Physical Human-Robot Interaction (Oct 2023 – now)

- Predict human intention using deep learning-based data-driven model.
- Develop adaptive reference generator to mitigate human-robot conflict.
- Devise semi-passive control to relax conservative of fully passive control and balance safe interaction and task performance.
- Analysis passivity, uniform ultimate boundedness, input-to-state stability.
- Implement human-in-the-loop experiments and analyze experiment data using statistical tests.

Adaptive Safety-critical Control for Uncertain MIMO Systems with Full-State Constraints and Input Saturation (Oct 2023 – now)

- Develop time-varying, asymmetric full-state constraints using barrier Lyapunov function control methods with log-type, tan-type, and universal-type.
- Relax boundedness assumptions and initial constrained values.
- Ensure input magnitude and input rate constraints.
- Analyze uniform ultimate boundedness and uniform asymptotic stability.

Adaptive Robust Finite-time/Fixed-time Control Approaches for a Class of Underactuated Mechanical Systems (Jan 2021 – now)

- Develop nonsingular hierarchical sliding mode control approaches to ensure convergence of both actuated and underactuated errors to zero.
- Resolve actuator faults, sensor faults, parametric uncertainties, and external disturbances by constructing nonlinear/ neural network/ fuzzy observers, extended state observers, and filters.
- Analyze practically finite-time/fixed-time stability.

Optimization-based Control Approaches for a Group of Multiple Surface Vehicles (Jan 2019 – Jan 2021)

- Develop a formation control scheme for multi-agent systems.
- Integrate formation control with an actor-critic reinforcement learning strategy.
- Analyze uniform ultimate boundedness and asymptotic stability.

WORK EXPERIENCE

Research Assistant, Human Robotics Lab, Nara Institute of Science and Technology

(Oct 2023 – now)

- Construct computational model to predict motion sickness symptoms and identify model parameters using optimization.
- Analyze experiment data and prediction data through statistical tests.

Teaching Assistant and Research Assistant, Mechatronics Engineering Lab, Hanoi University of Science and Technology

(Apr 2021 – Sep 2023)

- Manage, recruit, and train members of Mechatronics Engineering Lab.
- Complete the university project "Adaptive control for multi-axis drive systems with uncertain parameters and mismatched disturbances" with 01 ISI paper, 03 international conference papers.

Tutor in Student Learning Assistance Club, Hanoi University of Science and Technology

(Jan 2017 – Dec 2019)

- Support about 500 students in Mathematics (Calculus, Algebra, Probability and Statistics) and Physical.

PUBLICATION

- **V. T. Dang**, M. Bourseul, and T. Wada, "Prediction of individual motion sickness of bus passengers using a 6DoF-SVC model driven by handheld tablet motion during visual tasks," *IEEE Transactions on Intelligent Transportation Systems*, 2026, doi: 10.1109/TITS.2026.3710322.
- **V. T. Dang**, X. B. Nguyen, H. D. Pham, T. L. Nguyen, and V. T. Pham, "Adaptive finite-time hybrid tension–velocity control of uncertain multi-section web machines with uniform ultimate boundedness," *ISA Transactions*, vol. 170, pp. 200–216, 2026.
- V. B. Cao, **V. T. Dang**, V. T. Dam, and V. T. Pham, "Robust finite-time prescribed performance tracking control of autonomous surface vehicles without velocity measurement under time-varying actuator faults and ocean disturbances," *European Journal of Control*, vol. 89, 2026.
- **V. T. Dang**, T. D. T. Tran, D. B. H. Nguyen, and T. L. Nguyen, "Observer-based nonlinear cascade control approach of rewinding systems with uncertainties and disturbances compensation," *International Journal of Automation and Control*, vol. 20, no. 3, pp. 273–294, 2026.
- **V. T. Dang**, H. Kotake, S. Honji, T. Wada, "A Cooperation Control Framework Based on Admittance Control and Time-Varying Passive Velocity Field Control for Human–Robot Co-Carrying Tasks," in *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 23579-23593, 2025.
- V. T. Dang, D. L. Mac, D. T. Le, T. L. Nguyen, and **V. T. Dang**, "An online reinforcement learning-based conflict resolution approach for human–vehicle cooperative systems," *ISA Transactions*, vol. 169, pp. 391–407, 2025.
- S. Inoue, **V. T. Dang**, H. Liu and T. Wada, "Construction of a computational model of individual progression of motion sickness symptoms based on subjective vertical conflict theory," in *Experimental Brain Research*, vol. 243, pp. 108, 2025.
- H. T. Nguyen, **V. T. Dang**, T. M. Hoang, D. T. Le, D. M. Do, M. N. Vu and T. L. Nguyen, "A Fixed-Time Convergence Control of Roll-to-Roll Systems with a Fault-Tolerant Mechanism," in *IEEE Access*, vol. 13, pp. 138247-138264, 2025, doi: 10.1109/ACCESS.2025.3593826.
- **V. T. Dang**, X. B. Nguyen, T. D. T. Tran, D. T. Le and T. L. Nguyen, "Finite-time velocity sensorless integral sliding mode control for roll-to-roll systems under matched disturbances," in *ISA Transactions*, vol. 164, pp. 61-74, 2025.
- D. T. Le, M. T. Ngo, **V. T. Dang**, V. N. Giap, B. M. Nguyen, T. L. Nguyen and D. C. Hoang, "An Axial Gap Bearingless Motor Drive System with Nonlinear Disturbance Observers-Based Robust

Control," in *Journal of Control, Automation and Electrical Systems*, vol. 35, pp. 180–190, 2024, doi: 10.1007/s40313-023-01055-9.

- K. Nguyen, **V. T. Dang**, D. D. Pham and P. N. Dang, "Formation control scheme with reinforcement learning strategy for a group of multiple surface vehicles." in *International Journal of Robust and Nonlinear Control*, vol. 34.3, pp. 2252-2279, 2024, doi: 10.1002/rnc.7083.
- D. T. Le, M. T. Ngo, **V. T. Dang** and T. L. Nguyen, "A sensorless anti-windup speed control approach to axial gap bearingless motors with nonlinear lumped mismatched disturbance observers," in *ISA Transactions*, vol. 138, pp. 408-431, 2023, doi: 10.1016/j.isatra.2023.03.005.
- **V. T. Dang**, D. B. H. Nguyen, T. D. T. Tran, D. T. Le and T. L. Nguyen, "Model-free hierarchical control with fractional-order sliding surface for multisection web machines," in *International Journal of Adaptive Control and Signal Processing*, vol. 37, pp. 497-518, 2023, doi: 10.1002/acs.3534.

AWARDS & ACHIEVEMENTS

- Japanese Government (MEXT) Scholarship for P.h.D program, 2023-2026.
- Scholarship sponsored by Vingroup JSC and supported by the Master Scholarship Programme of Vingroup Innovation Foundation (VINIF), Institute of Big Data, Vietnam, 2022-2023.
- Certificate of Merit of President of Hanoi University of Science and Technology for the student who achieved an excellent classification degree, 2021.
- Talent scholarship of Hanoi University of Science and Technology, 2020.
- Certificate of Merit by the President of Hanoi University of Science and Technology in recognition of outstanding academic achievements and significant contributions to student initiatives, 2020.
- Scholarship to participate in Tokyo Tech AOTULE Summer Program, 2020.
- Scholarship of ABB corporation for the best student of the Department of Control Engineering and Automation, 2018-2020.

EXTRA CURRICULAR ACTIVITIES

Volunteer Manager, Sponsor a Student scholarship of Sunflower Mission in Hanoi (Oct 2020 – now)

- Support about 30 students who face financial hardship with high motivation and notable academic achievements.

Chairperson, Student Union of School of Electrical and Electronic Engineering, Hanoi University of Science and Technology (Oct 2018 – Oct 2019)

- Organize scientific research initiatives, academic competitions, and talent contests in the School and University.
- Manage the Youth Union and Student Association records for approximately 6,000 students.

SKILLS

Language: Full professional proficiency in English.

Programming: Proficiency in Matlab, Python, and ControlDesk.

Others: Leadership Time Management Problem Solving Data Analysis